

환자 1번에 대한 분석

High Risk Tumor vs Low Risk Tumor

	Gene	mean_high	mean_low	median_high	median_low	mean_diff	abs_mean_diff	cohen_d	abs_cohen_d	pvalue	fdr
0	CALML5	22.105384	24.258681	22.350606	24.418435	-2.153296	2.153296	-3.290279	3.290279	0.000006	0.042306
1	GCLM	9.983774	15.871834	10.025286	16.813172	-5.888059	5.888059	-2.347509	2.347509	0.000017	0.062242
2	FBXO6	9.231060	13.019191	9.022307	14.356788	-3.788131	3.788131	-2.039438	2.039438	0.000075	0.184566
3	PLA2G4A	9.135075	13.536460	8.749612	13.852249	-4.401385	4.401385	-2.134826	2.134826	0.000198	0.293195
4	MYH11	19.650839	17.666535	19.680930	17.764897	1.984304	1.984304	3.241305	3.241305	0.000168	0.293195
5	FBXO2	10.137622	15.470343	10.314749	17.998404	-5.332721	5.332721	-1.644138	1.644138	0.000542	0.340783
6	CTDNEP1	9.925646	15.158621	9.805765	17.561354	-5.232975	5.232975	-1.590661	1.590661	0.000728	0.340783
7	KIF22	15.628356	11.723128	15.532290	10.893160	3.905229	3.905229	1.594377	1.594377	0.000736	0.340783
8	EED	9.596078	13.486605	9.755687	14.912037	-3.890527	3.890527	-1.740028	1.740028	0.000333	0.340783
9	PTER	10.119024	13.295586	10.072416	14.084255	-3.176562	3.176562	-1.744881	1.744881	0.000546	0.340783
0	ACSL5	10.028825	13.119398	9.926467	14.333369	-3.090574	3.090574	-1.585215	1.585215	0.000726	0.340783
1	APOBEC3C	13.754879	10.955697	13.919059	10.662551	2.799182	2.799182	1.610106	1.610106	0.000783	0.340783
2	FLYWCH2	13.826090	11.681425	13.795074	12.233506	2.144665	2.144665	1.779097	1.779097	0.000644	0.340783
3	GOLM2	9.991453	11.949610	9.949414	12.521869	-1.958156	1.958156	-1.684772	1.684772	0.000432	0.340783
4	GSTK1	18.429985	19.959416	18.558736	20.253789	-1.529431	1.529431	-2.124550	2.124550	0.000692	0.340783
5	SLC35E2B	9.070567	10.447924	9.110466	10.609740	-1.377357	1.377357	-1.772557	1.772557	0.000747	0.340783
6	DTNA	10.714200	9.518283	10.742772	9.437513	1.195918	1.195918	2.208478	2.208478	0.000320	0.340783
7	MIA	15.584935	17.232256	15.719525	17.011064	-1.647321	1.647321	-1.644983	1.644983	0.000834	0.342852
8	ST3GAL6	9.647426	13.639547	10.137599	15.097136	-3.992121	3.992121	-1.605365	1.605365	0.001071	0.385285
9	CNN1	17.484991	14.742734	17.671850	15.247651	2.742257	2.742257	1.578018	1.578018	0.001035	0.385285

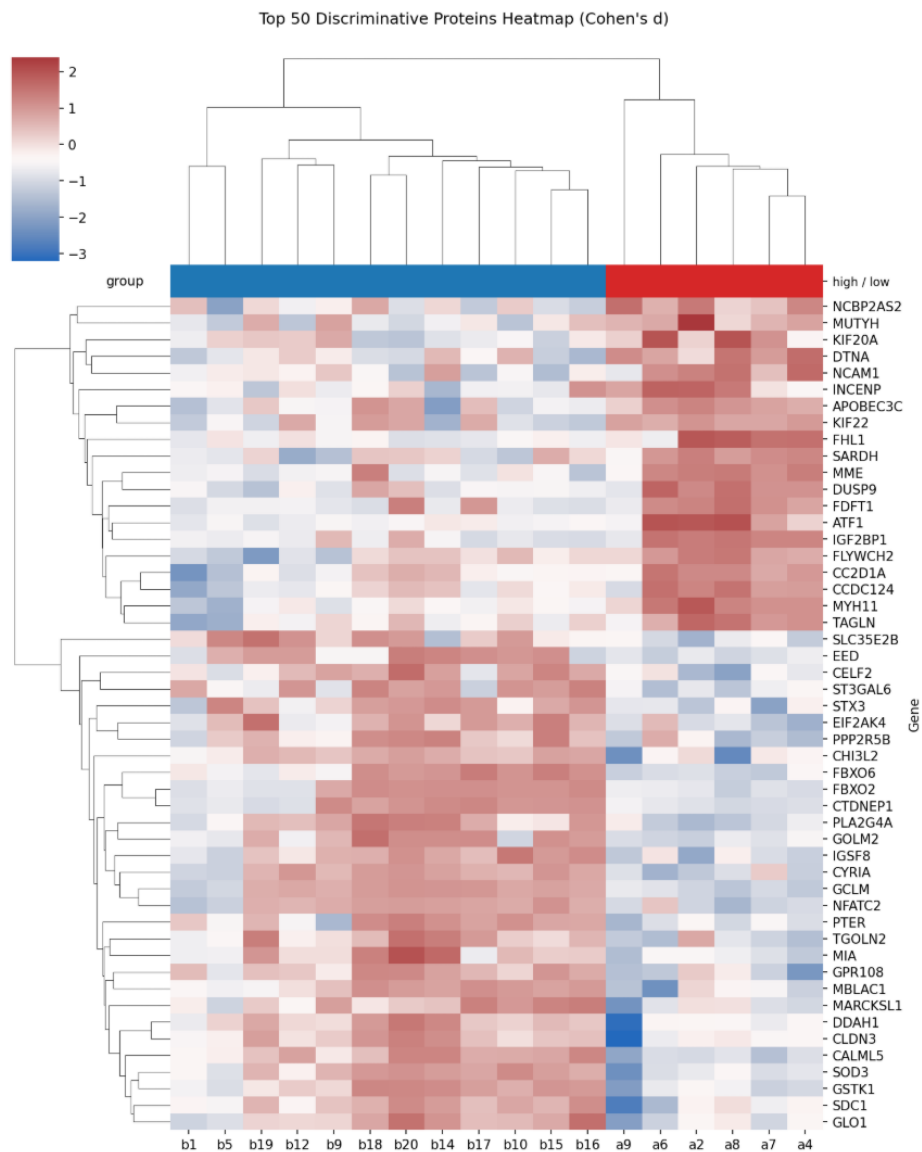
High Risk T-cell vs Low Risk T-cell

	Gene	mean_high_t_cell	mean_low_t_cell	median_high_t_cell	median_low_t_cell	mean_diff	abs_mean_diff	cohen_d	abs_cohen_d	pvalue	fdr
	MFAP4	9.707997	15.526910	9.744129	16.266768	-5.818913	5.818913	-3.245735	3.245735	0.000013	0.065394
	MYO5B	9.310715	13.380826	9.220295	13.642412	-4.070112	4.070112	-3.073329	3.073329	0.000027	0.065394
	NFIC	10.226854	14.253958	10.191255	14.672341	-4.027103	4.027103	-3.010691	3.010691	0.000025	0.065394
	MRPS15	9.377633	13.356533	9.457933	13.630304	-3.978899	3.978899	-2.866283	2.866283	0.000038	0.071002
	NUDCD2	10.464744	16.565880	10.346548	16.305458	-6.101136	6.101136	-6.678736	6.678736	0.000055	0.080733
	SLC66A3	10.287194	11.746098	10.244000	11.673553	-1.458904	1.458904	-2.550111	2.550111	0.000214	0.264228
	MMP14	10.063535	13.544309	10.082187	13.967143	-3.480774	3.480774	-2.183015	2.183015	0.000387	0.408793
	ACACA	9.626222	17.645069	9.371342	19.579197	-8.018847	8.018847	-2.089962	2.089962	0.000513	0.421713
	NR2C2AP	10.179609	12.784344	10.225162	12.814087	-2.604735	2.604735	-3.718172	3.718172	0.000498	0.421713
	PARG	9.822525	11.503471	9.969353	12.016273	-1.680945	1.680945	-2.004317	2.004317	0.000737	0.545116
	RHOC	9.626530	12.244595	9.919520	12.567981	-2.618066	2.618066	-2.369733	2.369733	0.000968	0.599571
	NOL7	9.086864	11.496006	8.905635	10.951636	-2.409142	2.409142	-1.929782	1.929782	0.000973	0.599571
	SERPINA6	9.266139	13.367006	9.134491	13.956368	-4.100867	4.100867	-2.229228	2.229228	0.001305	0.715243
	GALNS	9.475674	13.470398	9.507810	14.496884	-3.994724	3.994724	-1.795455	1.795455	0.001480	0.715243
	MYCBP	10.019691	13.555924	9.822930	14.642157	-3.536234	3.536234	-1.809144	1.809144	0.001459	0.715243
	B4GALT7	9.852132	11.921485	9.934855	11.996252	-2.069353	2.069353	-1.768447	1.768447	0.001723	0.715243
	TPD52L2	25.661221	27.643640	25.700643	27.338464	-1.982419	1.982419	-1.739826	1.739826	0.001837	0.715243
	VPS41	9.254396	11.055971	9.356799	11.153078	-1.801575	1.801575	-2.144058	2.144058	0.001760	0.715243
	HIGD1A	9.744776	10.972655	9.794935	11.129883	-1.227879	1.227879	-1.763618	1.763618	0.001691	0.715243
	BAG5	9.582194	15.245363	9.739036	17.516828	-5.663169	5.663169	-1.679499	1.679499	0.002398	0.777639

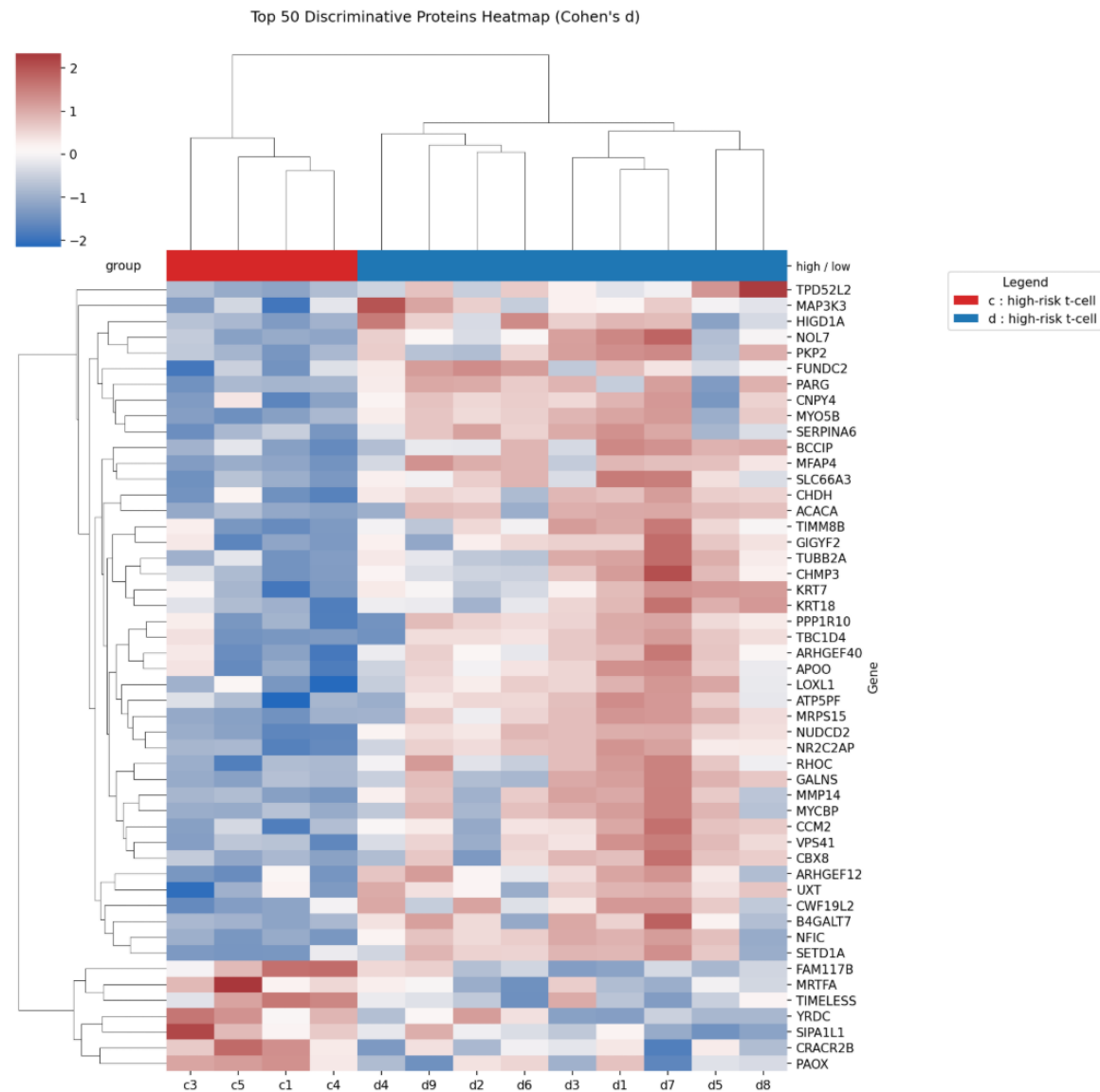
환자 1번에 대한 분석

Marker 중 p-value 0.05, abs_mean_diff > 0.58 이상인 마커들 시각화

High Risk Tumor vs Low Risk Tumor

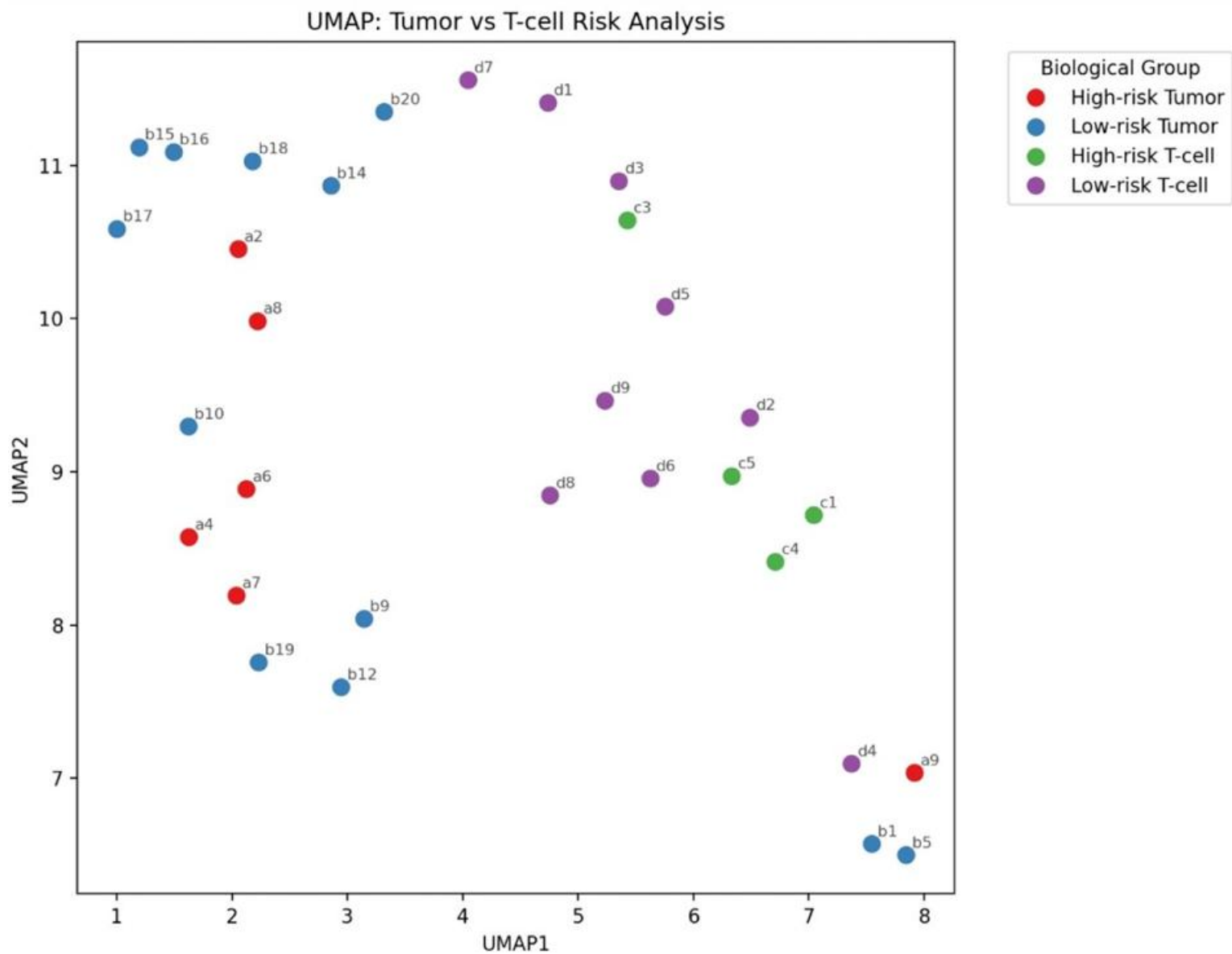


High Risk T-cell vs Low Risk T-cell



환자 1번에 대한 분석

전체 marker로 표현된 피처로 UMAP 시각화



환자 2번에 대한 분석

High Risk Tumor vs Low Risk Tumor

Gene	mean_high	mean_low	median_high	median_low	mean_diff	abs_mean_diff	cohen_d	abs_cohen_d	pvalue	fdr
ANK1	10.045246	14.199634	9.122392	15.482585	-4.154388	4.154388	-1.626094	1.626094	0.000349	0.916154
DPT	16.481959	20.468022	20.975513	21.916364	-3.986063	3.986063	-0.757912	0.757912	0.080066	0.916154
WSCD1	17.013250	13.220526	18.182235	10.419168	3.792724	3.792724	0.964325	0.964325	0.015399	0.916154
COL6A6	10.184113	13.717939	9.630527	15.471949	-3.533826	3.533826	-1.352833	1.352833	0.001296	0.916154
RFTN1	12.265841	15.779893	10.197075	17.875033	-3.514052	3.514052	-0.837349	0.837349	0.044863	0.916154
SP90AB2P	17.787299	14.393565	19.941231	10.837846	3.393734	3.393734	0.652820	0.652820	0.099298	0.916154
ZNF362	12.479570	15.752046	11.141979	17.428832	-3.272476	3.272476	-0.856256	0.856256	0.040373	0.916154
C8B	12.637551	15.842422	12.727072	15.980015	-3.204871	3.204871	-1.007676	1.007676	0.028542	0.916154
BNIP1	12.956123	16.125712	13.331718	16.566211	-3.169589	3.169589	-1.001679	1.001679	0.030208	0.916154
KRT31	17.306728	14.141889	17.957987	16.141942	3.164839	3.164839	0.784971	0.784971	0.051817	0.916154
COL6A3	36.202450	39.363031	37.842870	38.904120	-3.160581	3.160581	-0.930652	0.930652	0.046143	0.916154
ZNF511	14.146148	17.285872	14.396456	18.463448	-3.139724	3.139724	-0.813221	0.813221	0.059581	0.916154
FABP4	10.460876	13.457930	9.047730	15.410615	-2.997054	2.997054	-0.949251	0.949251	0.020510	0.916154
BAG1	12.780247	15.763323	10.623000	20.178980	-2.983075	2.983075	-0.561912	0.561912	0.156133	0.916154
POSTN	55.174987	58.151035	55.939385	58.314345	-2.976048	2.976048	-0.584361	0.584361	0.140486	0.916154
ADAR	31.377057	34.085153	30.125505	36.134935	-2.708097	2.708097	-0.822130	0.822130	0.044511	0.916154
ABL2	14.213383	16.873245	10.625024	19.438752	-2.659862	2.659862	-0.543171	0.543171	0.179819	0.916154
GBP6	12.130729	14.776560	10.118436	16.838140	-2.645831	2.645831	-0.661707	0.661707	0.102709	0.916154
CIR1	10.311886	12.888352	10.055046	13.820371	-2.576466	2.576466	-0.940613	0.940613	0.017001	0.916154
RBP5	12.110420	14.686209	12.290293	15.757992	-2.575790	2.575790	-0.899885	0.899885	0.034078	0.916154

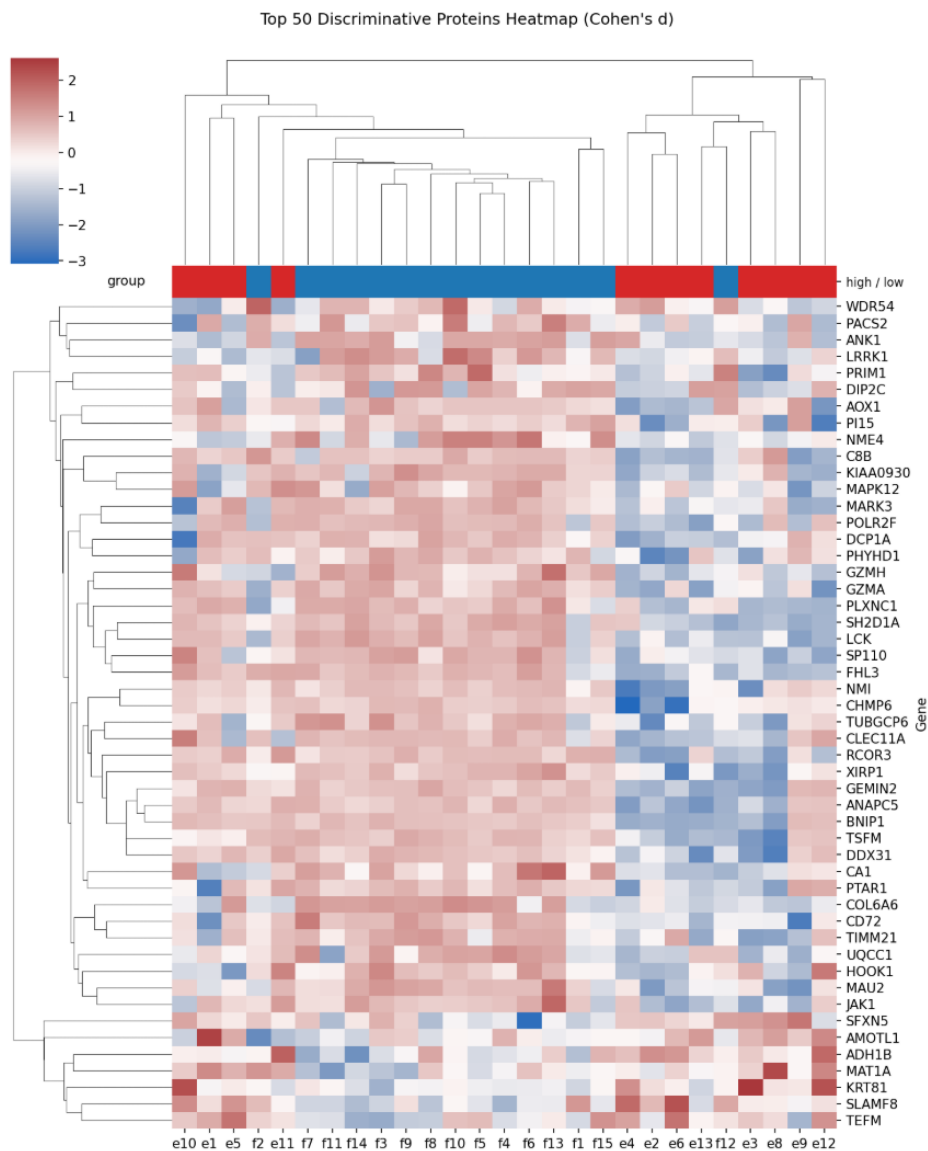
High Risk T-cell vs Low Risk T-cell

Gene	mean_high	mean_low	median_high	median_low	mean_diff	abs_mean_diff	cohen_d	abs_cohen_d	pvalue	fdr
TFAP2C	8.953307	10.467974	8.841044	10.442787	-1.514667	1.514667	-2.790660	2.790660	0.000114	0.840517
P3H2	9.922799	16.526276	9.796218	16.844117	-6.603477	6.603477	-1.239005	1.239005	0.036744	0.999595
LIME1	16.752727	10.924145	17.662866	9.858433	5.828582	5.828582	2.079155	2.079155	0.001364	0.999595
TUT4	9.535708	14.840194	9.896325	14.900093	-5.304486	5.304486	-1.321830	1.321830	0.026814	0.999595
CBS	11.648801	16.811273	9.590221	17.777498	-5.162473	5.162473	-1.421985	1.421985	0.026602	0.999595
CST4	16.042059	11.026980	16.601942	10.003153	5.015079	5.015079	1.205360	1.205360	0.049730	0.999595
USP13	11.752179	16.446283	10.150679	17.181967	-4.694104	4.694104	-1.545597	1.545597	0.017458	0.999595
MYL1	10.937114	15.390076	9.598582	16.003981	-4.452962	4.452962	-1.835282	1.835282	0.004853	0.999595
TK1	13.101180	17.501496	11.831533	17.709830	-4.400316	4.400316	-1.371171	1.371171	0.043804	0.999595
ARL6IP1	11.305229	15.694568	10.718662	18.101152	-4.389338	4.389338	-1.236332	1.236332	0.029946	0.999595
P4HA2	15.086443	19.439027	18.770095	19.507479	-4.352584	4.352584	-1.209611	1.209611	0.072032	0.999595
PFN2	31.644692	35.907855	34.340350	37.968677	-4.263163	4.263163	-0.803681	0.803681	0.149227	0.999595
IGF2BP1	11.891960	16.035277	10.993584	16.542178	-4.143317	4.143317	-1.292154	1.292154	0.037396	0.999595
CSPG4	13.341174	17.386944	17.376864	17.389454	-4.045769	4.045769	-1.161760	1.161760	0.081312	0.999595
PACSIN3	17.009212	20.848110	21.880981	22.157530	-3.838898	3.838898	-0.702447	0.702447	0.219120	0.999595
CLIC6	16.191029	12.510820	18.763891	10.547205	3.680208	3.680208	0.745424	0.745424	0.179481	0.999595
ERI3	9.316437	12.929893	9.603950	14.505775	-3.613456	3.613456	-2.007984	2.007984	0.002847	0.999595
EIF4EBP1	10.923514	14.535009	10.319329	15.411620	-3.611495	3.611495	-1.451545	1.451545	0.018200	0.999595
GSTA1	13.249784	16.824696	10.539417	16.505641	-3.574911	3.574911	-1.160841	1.160841	0.072211	0.999595
COL11A1	13.039951	16.562293	10.380763	17.943893	-3.522342	3.522342	-0.871709	0.871709	0.114620	0.999595

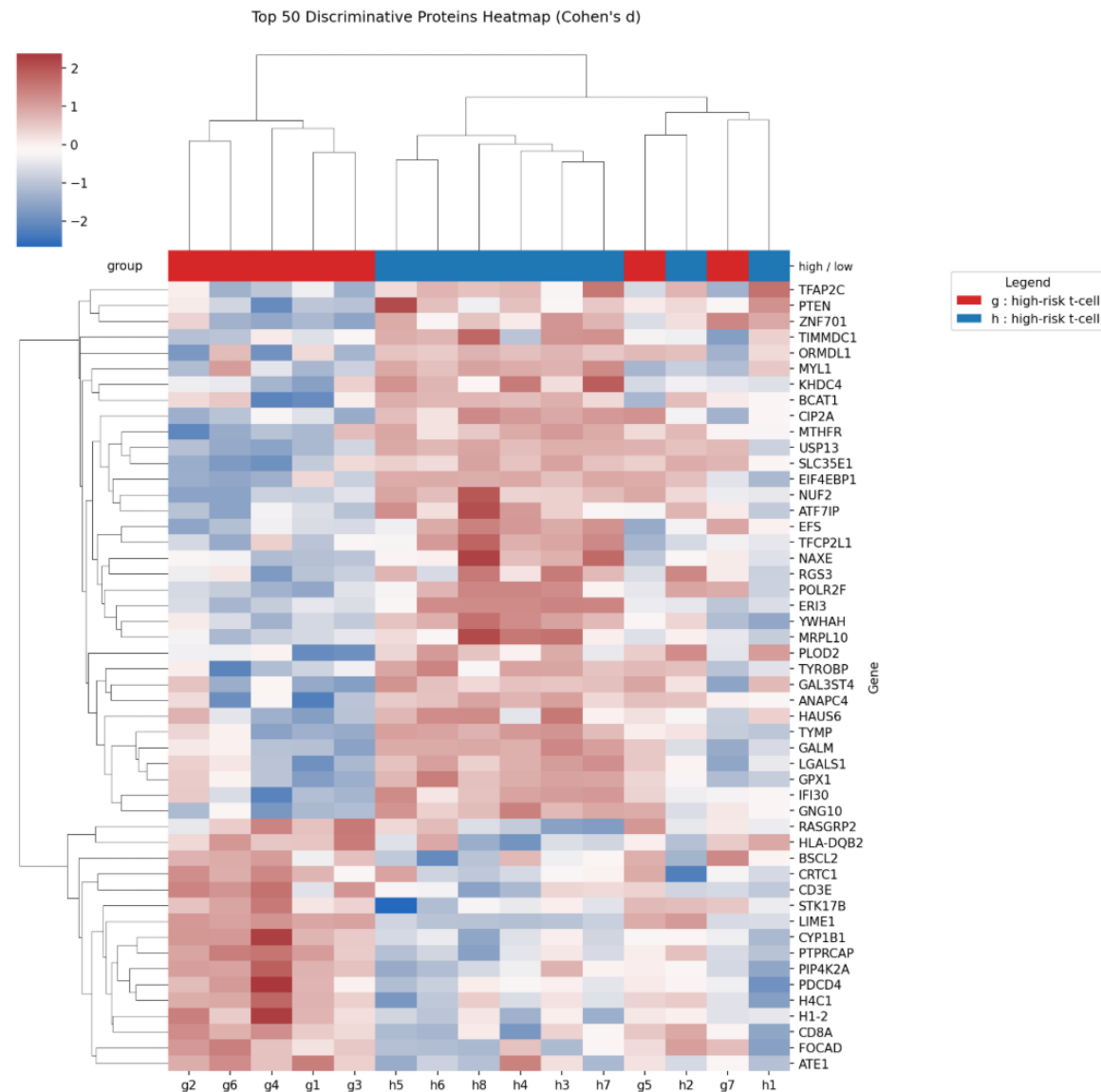
환자 2번에 대한 분석

Marker 중 p-value 0.05, abs_mean_diff > 0.58 이상인 마커들 시각화

High Risk Tumor vs Low Risk Tumor

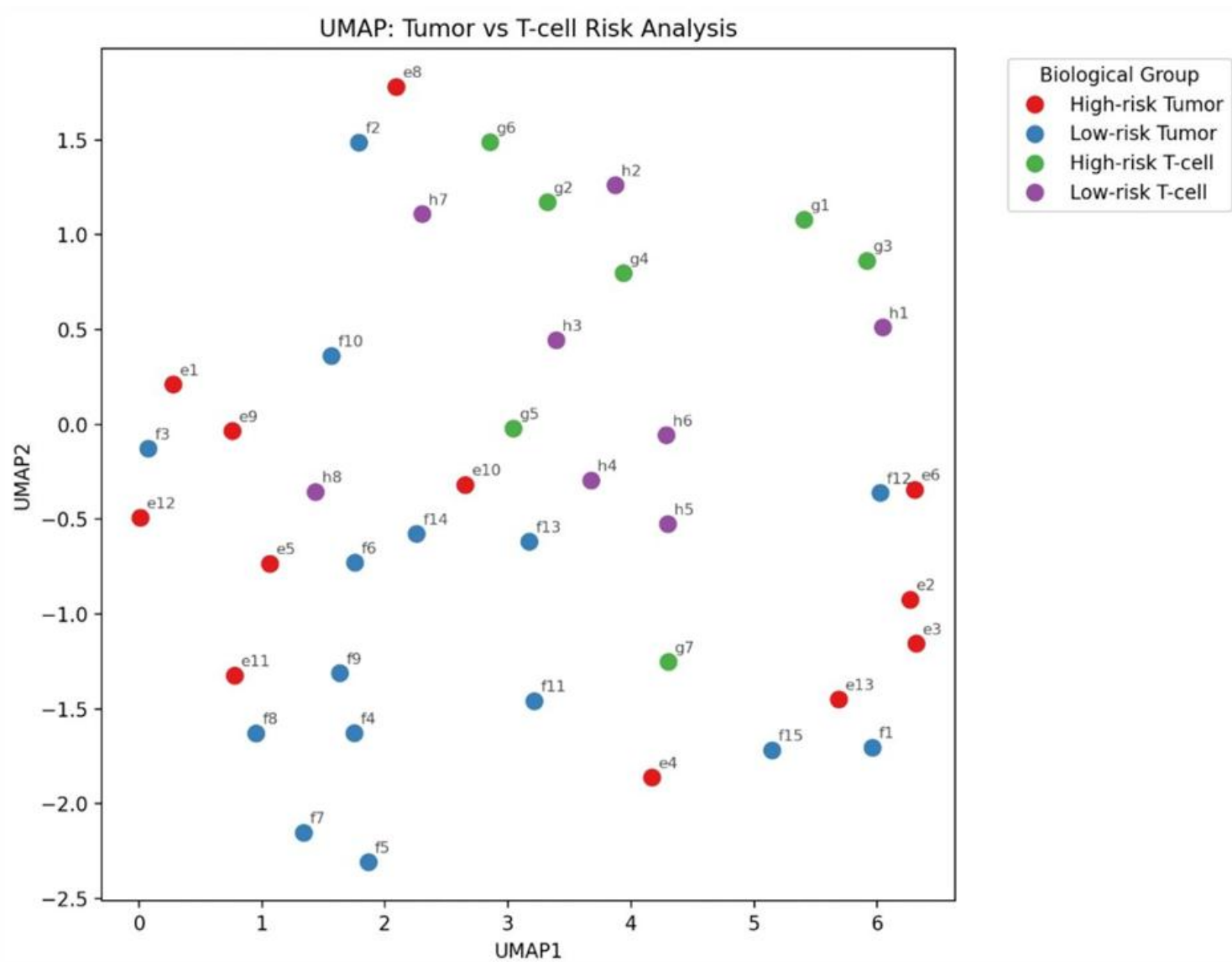


High Risk T-cell vs Low Risk T-cell



환자 2번에 대한 분석

전체 marker로 표현된 피쳐로 UMAP 시각화



환자 2명에 대해 공통적으로 차이가 나타난 Marker 들

Tumor 영역

	Mean diff (high – low) 1	Mean diff (high – low) 2	Category
DTNA	1.195	0.904	High
FABP4	-2.145	-2.997	Low
IRF7	-1.126	-1.219	Low
MANB A	-1.859	-1.267	Low
RBM23	0.889	-0.860	Unmatched
SPG21	-1.327	-0.890	Low
USP49	2.808	-1.432	Unmatched

T-cell 영역

	Mean diff (high – low) 1	Mean diff (high – low) 2	Category
ADI1	-3.783	-0.750	Low
ATN1	-1.262	-1.377	Low
BCAT1	-2.220	-2.865	Low
CACNA2 D1	-2.083	1.867	Unmatched
EIF4ENIF1	-0.706	-0.8553	Low
NUF2	-1.014	-1.035	Low
RHPN2	-1.556	-1.785	Low
TRIM27	-2.445	-2.995	Low
UBR3	-1.558	-1.977	Low